

V2V-Based Cooperative Control of Heterogeneous CAV Platoons: An Intelligent VO-IDA Approach

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Abstract—To overcome the heterogeneous dynamics and unreliable communication which adversely effect the stable control for connected and automated vehicle platoons, a new intelligent control approach, named virtual order-degradation interconnection and damping assignment (VO-IDA), is proposed in this article. First, the internal stability of vehicle platoons is abstracted into a class of tracking control problems for general chained integral systems. By converting the chained integral system into standard closed-loop port-controlled Hamiltonian form, VO-IDA achieves asymptotic tracking through the integration of backstepping order degradation and virtual stabilization control techniques. This conversion effectively eliminates the dependence on preceding vehicular acceleration as well. Second, under heterogeneous dynamics, explicit stable domains of control parameters are provided to ensure the attenuation of string stability for vehicle platoons via Laplace transform. Furthermore, a linear-proportional relationship between heterogeneous and homogeneous dynamics regarding spacing error ratio is uncovered. Leveraging this relationship, a modified multiobjective genetic algorithm is employed to online explore target locations within stable domains, enabling VO-IDA to conduct stable and precise control under heterogeneous dynamics. Comparative experiments verify the superiority of this approach.

Index Terms—Connected vehicle, heterogeneous platoon, passivity-based control, port-controlled Hamiltonian (PCH), string stability.

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I. INTRODUCTION

THE VEHICLE-TO-VEHICLE (V2V) communication is evolving rapidly. Relying upon this technique, vehicles traveling on the road are transformed from islands into intelligent terminals, enabling fast, accurate and diverse information sharing with each other [1]. As a typical application of V2V, the cooperative control of connected and automated vehicle (CAV) platoons is becoming a promising research focus with its potential to alleviate traffic congestion, guarantee driving safety, enhance riding comfort and reduce energy emission [2]. Nevertheless, this advancement also brings about heightened complexity in addressing related problems, necessitating advanced control techniques to effectively manage them.

The most essential objective of CAV platoons control is to guarantee internal stability and string stability, the former requires the follower vehicles to rapidly and accurately track the desired interspacing and the steady velocity of the leader vehicle simultaneously; the latter necessitates the disturbances generated from the platoon header to have an attenuation property on tracking accuracy as propagating backward along this platoon, which effectively avoids stop-and-go pattern and rear-end collision caused by drastic motion variation of the leader vehicle [3]. A comprehensive review on string stability is provided in [4].

The stability properties are closely related to the spacing policy, vehicular dynamics and communication reliability.

Spacing Policy Type: The earlier research predominantly employed constant distance policy, which is struggled to guarantee string stability using linear feedback controllers under distributed communication topologies [5]. The constant time headway (CTH) policy proposed in [6] establishes a linear relationship between desired intervehicle spacing and velocity, which has been widely adopted subsequently as it aligns more closely with driver characteristics and guarantee string stability in distributed topologies [7].

Vehicular Dynamics: In order to simplify the controller design, many previous approaches assume that all the vehicles have homogeneous dynamics [8], [9]. However, heterogeneous dynamics are inevitably encountered in practice. Due to the net-connected mode, each vehicle among the platoon is highly coupled, rendering the approaches relying on the homogeneous assumption struggled to ensure substantial stability in heterogeneous case. In [10], the linear transformation and eigenvalue decomposition have been introduced to transform the coupled relationship controller matrix into uncertain part

of the model, with a robust controller designed to handle heterogeneity. This type of approaches divides the vehicular dynamic model into nominal homogeneous and uncertain heterogeneous parts, aiming to guarantee robustness under the bounded perturbation by strategies, such as robust control, sliding mode control (SMC), [11], [12], [13], etc. In [14], a nominal adaptive SMC has been employed to ensure that the desired vehicle acceleration is achieved when the designed sliding mode surface converges to zero as a way to alleviate the adverse effects of heterogeneous dynamics. This type of approaches considers the nonnominal heterogeneous part of the model as an unknown term, which is compensated by active disturbance rejection control or adaptive control techniques [15], [16]. In [17], a distributed model predictive control approach has been proposed for heterogeneous vehicle platoons, incorporating a zero terminal equality constraint in cost-function solving process to progressively correct the position errors. This type of approaches compensates the error arising from heterogeneous dynamics by constraint design in time-domain objectives solving process [18], [19]. On the flip side, the strong frequency-domain string stability (SFSS) [20] has been widely used for string stability analysis upon the linear vehicular system with homogeneous dynamics, as it can easily guarantee the attention property [21]. In comparison to the approaches above that treat the heterogeneous part as model uncertainty, unknown terms, or incorporate constraints into optimization, SFSS allows an explicit characterization for the impact of heterogeneous dynamics on system performance during its analysis process [22]. This advantage mitigates the conservativeness in string stability analysis, allowing the corresponding string stability criterion to determine a larger feasible domain. However, most of the existing SFSS analyses are limited to homogeneous dynamics, since heterogeneous dynamics render the analysis more challenging [23].

Communication Reliability: Unreliable communication (time delay, packet loss, and channel fading) encountered in practice has been found to have a significant impact on string stability [24]. The preceding vehicular acceleration (a_{i-1}) is relied upon by a large portion of CAV platoon controllers, though it is sensitive to communication failures. One type of approaches aims to alleviate the adverse effects caused by unreliable a_{i-1} [25], [26]. To liberate controllers from relying on a_{i-1} , a disturbance observer (DO) has been introduced in [15] to estimate a_{i-1} and a heterogeneous vehicular dynamic part, which is then incorporated into SMC. In [27], an integral sliding mode surface has been proposed, which no longer requires the controller to rely on a_{i-1} , along with an extended DO to directly estimate multiple disturbances.

On the other hand, a large number of physical systems in the real world can be modeled and analyzed by port-Hamiltonian (PH) systems [28]. The port-controlled Hamiltonian (PCH) systems with dissipation, as a significant special case of PH, have received widespread attention in control field due to its advantages of high efficiency and high-energy density in characterizing the system energy [29], [30]. The interconnection damping assignment passivity-based control (IDA-PBC) has been proposed by Ortega et al. [31] for PCH systems, which is one of the most powerful approaches in passive control and

is widely used in industrial applications [32], [33]. Based on dissipation theory, a controller is designed by constructing the input-output mapping relations, ensuring that the closed-loop system satisfies the passive condition for stabilization. Many extended researches upon IDA-PBC have emerged in recent years [34], [35], [36]. In [34], an integral action (IA) approach superimposed on IDA-PBC has been proposed, which adds an integral feedback term while preserves the PCH form. Similar to what the common IA roles, this approach further improves the control performance of IDA-PBC, such as enhancing tracking accuracy, alleviating unmatched disturbance, and eliminating static errors.

There is scarce research, to the author's knowledge, on how to apply PCH-based approaches to the cooperative control of CAV platoons while exploit their potential to ensure stability and address issues, such as heterogeneous dynamics and unreliable communication. With these aims, this article proposes a new PCH-based control approach, called virtual order-degradation interconnection and damping assignment (VO-IDA). Combined with intelligent algorithm, the VO-IDA is applicable for the stable control of CAV platoons with heterogeneous dynamics and unreliable a_{i-1} . Compared with relevant works, this article makes threefold contributions.

- 1) A VO-IDA approach is first proposed for the chained integral systems featuring two higher relative-degree (HRD) and one relative-degree-one (RD1) channels. It addresses the technical challenge of directly applying IDA-PBC and IA to such kind of systems by introducing backstepping technique for order degradation and combining virtual stabilization technique with IDA-PBC to eliminate nonstandard terms. Subsequently, the proposed VO-IDA is implemented within longitudinal control for CAV platoons to ensure internal stability.
- 2) For the unreliable a_{i-1} , in contrast to the ways involving alleviating the corresponding adverse effects or estimating a_{i-1} , the virtual stabilization in VO-IDA indirectly releases the dependence on a_{i-1} for the controller without compromising control performance, such as control accuracy. This fundamentally avoids the adverse effects arising from unreliable a_{i-1} .
- 3) An SFSS analysis is provided, explicitly considering heterogeneous dynamics, to establish a feasible domain (stable domain) with less conservatism. On the flip side, upon employing the VO-IDA controller, a linear-proportional relationship upon acceleration and spacing error ratio under heterogeneous dynamics is revealed, which provides a basis for exploring the target locations within the corresponding stable domains. This exploration is online achieved by a modified multiobjective genetic algorithm (MOGA), empowering the VO-IDA controller to handle dynamical heterogeneity, and further enhancing the attenuation property upon string stability.

The remainder of this article is organized as follows. Section II clarifies the motivations of this article. The VO-IDA approach is proposed in Section III to guarantee internal stability. Section IV analyzes the SFSS and then details an online exploration of target locations within stable domains to achieve

string stability. Comparative experiments are conducted in Section V and finally a conclusion is made in Section VI.

II. PROBLEM FORMULATION AND PRELIMINARIES

Consider a platoon of CAVs with index 0 for the leader vehicle and 1 to N for the follower ones. The longitudinal vehicle dynamic system for the i th follower is given by

$$\dot{s}_i = v_i \quad (1a)$$

$$\dot{v}_i = a_i \quad (1b)$$

$$\dot{a}_i = -\frac{1}{\tau_i}a_i + \frac{1}{\tau_i}u_i \quad (1c)$$

where s_i , v_i , and a_i denote the position, velocity and acceleration, respectively, and u_i is the controller to be designed, τ_i is the first-order lumped time lag which mainly depends on the process of engine response and throttle/brake actuator in powertrain [37]. In practice, τ_i among the follower vehicles is inevitably different, reflecting the dynamical heterogeneity.

To be emphasized here, the system (1) does not originally possess a PCH structure. For systems without this structure, the IDA-PBC requires a controller to reconstruct the original system into a closed-loop PCH form. This involves the design and solution of unknown partial differential equations (PDEs), the number of which is especially much greater for higher order systems, making it technically difficult to solve [35].

The CTH spacing policy is employed in this article

$$d_i = h_i v_i + d_{m,i} \quad (2)$$

where d_i represents the desired distance between the i th follower vehicle and its reference one, h_i is time headway, and $d_{m,i}$ is the sum of a standstill distance and the length of vehicle $i - 1$. Thus, the spacing error is defined as

$$e_{1,i} = s_i - s_{i-1} + h_i v_i + d_{m,i}. \quad (3)$$

Now, we present the following definitions, lemmas, and an assumption for later development.

Definition 1 [34]: Give a system as below

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_h \end{bmatrix} = \begin{bmatrix} A_1 & A_{1h} \\ A_{h1} & A_h \end{bmatrix} \begin{bmatrix} x_1 \\ x_h \end{bmatrix} + \begin{bmatrix} g_1 \\ 0 \end{bmatrix} u \quad (4)$$

where the matrix g_1 has full row rank. The n_1 states x_1 receiving a direct control action are called RD1 states, corresponding to RD1 channels. The n_h remaining states x_h are called HRD states, corresponding to HRD channels.

Definition 2 ([4] SFSS): For the linear vehicle platoon systems with the preceding following (PF) topology, SFSS holds if the transfer function of spacing errors between follower $i - 1$ and i denoted as $G_{i-1,i}(j\omega)$, is such that [4]

$$\|G_{i-1,i}(j\omega)\|_{\mathcal{H}_\infty} \leq 1, \quad (\forall \omega > 0, i = 2 \cdots N) \quad (5)$$

where $\forall \omega > 0$ denotes frequency, j denotes an imaginary number, and $G_{i-1,i}(s)$ denotes the transfer function between spacing errors $e_{1,i-1}$ and $e_{1,i}$.

Definition 3 (Consistency): Subject to the systems and topology in Definition 2, on the basis of SFSS, the consistency property indicates the following relationship:

$$\frac{a_i(s)}{e_{1,i}(s)} = \frac{a_1(s)}{e_{1,1}(s) \prod_{q=2}^i \eta_q}, \quad (i = 2, \dots, N) \quad (6)$$

where η_q represents the attenuation factor of the q th follower, the accelerations a_i and spacing errors $e_{1,i}$ undergo the Laplace transform. Similar to the analysis for SFSS, the lumped time lag is ignored here.

Consistency indicates that, given the assurance of internal stability, when there are fluctuations one after the other in the motion behavior, the trend of vehicular spacing errors remains consistent, specifically the trend of signs. Otherwise, despite a attenuation in the error values, the intervehicle distance may still get closer, which is detrimental to driving safety [38].

Lemma 1 [39]: Given a casual-stable system $\zeta_i(s)$, with its $\mathcal{H}_\infty$ norm defined as

$$\|\zeta_i(s)\|_{\mathcal{H}_\infty} = \sup_{\text{Re}(s)>0} \bar{\sigma}(P_i(s)) \quad (7)$$

where $\bar{\sigma}(P_i(s))$ is the maximum singular value.

The $\mathcal{H}_\infty$ norm of $\zeta_i(s)$ is calculated by evaluating along the imaginary, that is

$$\sup_{\text{Re}(s)>0} \bar{\sigma}(\zeta_i(s)) = \sup_{\text{Re}(s)>0} \bar{\sigma}(\zeta_i(j\omega)). \quad (8)$$

Consequently, the terms s and $j\omega$ are employed interchangeably in this article.

Besides, there is a relation between $\|\zeta_i(s)\|_{\mathcal{H}_\infty}$, input $\zeta_{\text{In},i}(t)$ and output $\zeta_{\text{Out},i}(t)$ as

$$\|\zeta_i(s)\|_{\mathcal{H}_\infty} = \sup_{\zeta_{\text{In},i} \neq 0} \frac{\|\zeta_{\text{Out},i}(t)\|_{\mathcal{L}_2}}{\|\zeta_{\text{In},i}(t)\|_{\mathcal{L}_2}}, \quad t \in (0, \infty) \quad (9)$$

where $\|\psi(t)\|_{\mathcal{L}_2} = \sqrt{\int_0^\infty \psi^T(t)\psi(t)dt} < \infty$ denotes the $\mathcal{L}_2$ norm for a piecewise continuous and square-integrable function $\psi(t)$.

Lemma 2 [40]: Define a state space $x \in \mathbb{R}^n$, an input spaces $u \in \mathbb{R}^m$ and an output spaces $y \in \mathbb{R}^m$. Consider a linear passive system (without feedthrough terms) as follows:

$$\dot{x} = Ax + Bu \quad (10a)$$

$$y = Cx \quad (10b)$$

and an input-state-output PCH system as follows:

$$\dot{x} = [J(x) - R(x)] \frac{\partial H}{\partial x}(x) + g_u(x)u \quad (11a)$$

$$y = g_u^T(x) \frac{\partial H}{\partial x}(x) \quad (11b)$$

where a Hamiltonian function $H : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as $(1/2)x^T Q x$, which represents power storage and is commonly regarded as the Lyapunov function. The n -order interconnection matrix $J(x)$ and damping matrix $R(x)$, respectively, represent power-conserving and internal dissipation of energy. The $g_u(x)$ is an input matrix.

Suppose that matrices $J(x)$, $R(x)$ and Q satisfy

$$J(x) = -J^T(x) \neq 0, \quad R(x) = R^T(x) > 0 \quad (12a)$$

$$A^T Q + Q A \leq 0, \quad B^T Q = C, \quad A Q^{-1} = J - R \quad (12b)$$

then the system (10) can be rewritten as the system (11).

Assumption 1: Multiple pieces of information are available among CAVs through V2V communication [41], [42], including controller parameters and the constant level of τ_i .

This research considers two control objectives for the cooperative control of CAV platoons.

- 1) *Internal Stability*: Determined by (3), the desired position for the i th follower is given as

$$s_i^* = s_{i-1} - h_i v_i - d_{m,i}. \quad (13)$$

The spacing error asymptotically converges to 0, when the leader is in a steady motion state

$$\lim_{t \rightarrow \infty} |s_i - s_i^*| = \lim_{t \rightarrow \infty} |e_{1,i}| = 0. \quad (14)$$

- 2) *SFSS*: The attenuation (5) and consistency (6) properties upon string stability are guaranteed simultaneously under homogeneous/heterogeneous dynamics.

III. DISTRIBUTED VO-IDA CONTROLLER DESIGN

In addressing the cooperative control problem upon CAV platoons, the system (1) and CTH policy (2) are utilized. The control objective for each follower is to ensure internal stability and SFSS simultaneously. In this section, only the internal stability is focused on, which serves as the basis for further SFSS in Section IV. In fact, this platoon control problem on the aspect of internal stability can be abstracted as a class of general control problems.

Consider a system

$$\dot{x}_1 = x_2 \quad (15a)$$

$$\dot{x}_2 = x_3 \quad (15b)$$

$$\dot{x}_3 = B(t)u + F(x_1, x_2, x_3, t) \quad (15c)$$

where the HRD channel exhibits a chain of integration characteristic, the RD1 channel features a linear input relationship, and $F(x_1, x_2, x_3, t)$ is a linear or nonlinear term composed of states. The control objective is to achieve an accurate tracking of the reference trajectory x_1^* for state x_1 , that is

$$x_1 \rightarrow x_1^*. \quad (16)$$

Noticing that (15) is a 3-order system without PCH structure, designing PDEs becomes challenging, as mentioned in Section II. On the other hand, a PCH-based IA approach has been verified to possess strong potential for enhancing control performance [34]. However, it is not theoretically applicable to PCH systems with more than one HRD channels. Therefore, to solve the trajectory tracking problem while avoiding complex PDEs design in IDA-PBC and being able to apply IA, a VO-IDA approach is proposed through Theorem 1.

Theorem 1: Consider the system (15) with a reference trajectory (16) for state x_1 . If there exist parameters $k, c, r, K, I, L, K_I(t), n(t)$ and $j(t)$, along with $B(t)$ in (15), satisfying the following conditions:

$$k > 0, c > 0, r > 0, K > 0, I > 0 \quad (17a)$$

$$n(t) > 0, L \neq 0, (L < 0 \Rightarrow n(t) \neq 1) \quad (17b)$$

$$(1 + \text{sgn}(L)n(t)) \text{sgn}(B(t))B(t) + j(t)c \neq 0 \quad (17c)$$

where $(\text{cond}_1 \Rightarrow \text{cond}_2)$ denotes that while condition cond_1 occurs, condition cond_2 ought to be correspondingly held.

Then the state x_1 asymptotically converges to the trajectory (16) by utilizing a VO-IDA controller as follows:

$$\begin{aligned} u = & - \left(\frac{I}{B(t)} + \frac{rc}{\text{sgn}(B(t))B^2(t)} \right) (x_3 - x_3^*) \\ & - \left(\frac{1 + \text{sgn}(L)n(t)}{B(t)} + \frac{j(t)c}{\text{sgn}(B(t))B^2(t)} \right. \\ & \quad \left. + \frac{K_I(t)K \text{sgn}(B(t))}{(1 + \text{sgn}(L)n(t)) \text{sgn}(B(t))B(t) + j(t)c} \right) (x_2 - x_2^*) \\ & - \frac{(\text{sgn}(B(t))B(t)I + rc) K_I(t)^2 K \text{sgn}(B(t))}{(1 + \text{sgn}(L)n(t))B^2(t) + \text{sgn}(B(t))B(t)j(t)c} \int (x_2 - x_2^*) dt \\ & - \frac{F(x_1, x_2, x_3, t)}{B(t)} \end{aligned} \quad (18)$$

where

$$x_3^* = \frac{\text{sgn}(L) \text{sgn}(B(t))B(t)n(t)(-x_2 + x_2^* + Lx_3) + j(t)cLx_3}{(1 + \text{sgn}(L)n(t)) \text{sgn}(B(t))B(t)L + j(t)cL} \quad (19)$$

$$x_2^* = -kx_1 + kx_1^* + \dot{x}_1^*. \quad (20)$$

Proof: Stage 1: In this stage, based on backstepping technique, the 3-order system (15) is degraded to a 2-order system, preliminarily possessing a PCH structure.

Define the coordinate variable

$$e_1 = x_1 - x_1^*. \quad (21)$$

Differentiate (21) along (15a)

$$\dot{e}_1 = x_2 - \dot{x}_1^*. \quad (22)$$

The Lyapunov function with respect to x_1 is

$$V = \frac{1}{2} e_1^2. \quad (23)$$

Differentiate (23) along (22)

$$\dot{V} = e_1(x_2 - \dot{x}_1^*). \quad (24)$$

Let the new target x_2^* of x_2 satisfy

$$x_2^* = \dot{x}_1^* - ke_1. \quad (25)$$

Combining (16), (21), and (25), then (20) is derived.

Define the coordinate variable

$$e_2 = x_2 - x_2^*. \quad (26)$$

Then combine (22), (25), and (26), we obtain

$$\dot{e}_1 = e_2 + \alpha_1 \quad (27a)$$

$$\alpha_1 = -ke_1 \quad (27b)$$

where k is the backstepping gain, $k > 0$. In this way, α_1 forms a virtual control over system (22), which is input-to-state stable if e_2 asymptotically converges to 0. Thus, the following proof only considers states x_2 and x_3 .

Combine (22) with (27)

$$-x_2 + Lx_3 = ke_1 - e_2 - \dot{x}_1^* + Lx_3. \quad (28)$$

Substituting (15b) and then introducing an auxiliary variable $\text{sgn}(L)n(t)$, (28) is reformulated as

$$\dot{x}_2 = -\frac{\text{sgn}(L)n(t)}{L}x_2 + (1 + \text{sgn}(L)n(t))x_3 - \frac{\text{sgn}(L)n(t)}{L}(ke_1 - e_2 - \dot{x}_1^* + Lx_3), \quad (L \neq 0) \quad (29)$$

where the coefficient of x_2 is designed to ensure the positive-definite property of the damping matrix in (12a).

By referencing (29), (15c) is reformulated as

$$\dot{x}_3 = -(1 + \text{sgn}(L)n(t))x_2 - Ix_3 + B(t)u + ((1 + \text{sgn}(L)n(t))x_2 + F(x_1, x_2, x_3, t) + Ix_3) \quad (30)$$

where $I > 0$. According to (12a), the coefficient of x_2 is designed to ensure the skew-symmetry property of the interconnection matrix, while the coefficient of x_3 is designed to ensure the positive-definite property of the damping matrix. Set an output equation for system (15)

$$y = \text{diag}(c, c)x_{23} \quad (31)$$

where $c > 0$ and $x_{23} = [x_2, x_3]^T$.

Define a Hamiltonian function

$$H(x_{23}) = \frac{c}{2 \text{sgn}(B(t))B(t)}x_2^2 + \frac{c}{2 \text{sgn}(B(t))B(t)}x_3^2. \quad (32)$$

Combining (29)–(32) and the relationship (12b) in Lemma 2, a system in the form of (11) is obtained as follows:

$$\dot{x}_{23} = [J - R] \frac{\partial H}{\partial x_{23}}(x_{23}) + g_u \begin{bmatrix} 0 \\ u \end{bmatrix} + \Xi \quad (33)$$

where

$$J = \begin{bmatrix} 0 & \frac{(1 + \text{sgn}(L)n(t)) \text{sgn}(B(t))B(t)}{c} \\ -\frac{(1 + \text{sgn}(L)n(t)) \text{sgn}(B(t))B(t)}{c} & 0 \end{bmatrix} \quad (34)$$

$$R = \begin{bmatrix} \frac{\text{sgn}(L) \text{sgn}(B(t))B(t)n(t)}{cL} & 0 \\ 0 & \frac{\text{sgn}(B(t))B(t)I}{c} \end{bmatrix} \quad (35)$$

$$g_u = \begin{bmatrix} B(t) & 0 \\ 0 & B(t) \end{bmatrix} \quad (36)$$

$$u = u_R + u_I \quad (37)$$

$$\Xi = \begin{bmatrix} -\frac{\text{sgn}(L)n(t)}{L}(ke_1 - e_2 - \dot{x}_1^* + Lx_3) \\ (1 + \text{sgn}(L)n(t))x_2 + F(x_1, x_2, x_3, t) + Ix_3 \end{bmatrix}. \quad (38)$$

Here, it is explained that the controller u is divided into IDA-PBC u_R and IA u_I parts, which are designed in stage 2 and stage 3, respectively. The $\Xi = [\Xi_1, \Xi_2]^T$ is a nonstandard term with regard to the PCH form.

Stage 2: In this stage, system (33) is stabilized as a standard closed-loop PCH system. The Ξ_1 and Ξ_2 are eliminated by virtual stabilization and IDA-PBC techniques, respectively.

Set J_a , R_a and H_K as follows:

$$J_a = \begin{bmatrix} 0 & j(t) \\ -j(t) & 0 \end{bmatrix} \quad (39)$$

$$R_a = \text{diag}(0, r) \quad (40)$$

$$H_K(x_{23}) = \frac{c}{2 \text{sgn}(B(t))B(t)}(-2x_2^*x_2 + x_2^{*2}) + \frac{c}{2 \text{sgn}(B(t))B(t)}(-2x_3^*x_3 + x_3^{*2}) \quad (41)$$

where x_3^* is to be determined, $r > 0$.

Combining (34) and (39), (35) and (40), (32) and (41), respectively, we have

$$J_d = J + J_a = -J_d^T \quad (42)$$

$$R_d = R + R_a = R_d^T \quad (43)$$

$$H_d(x_{23}) = H(x_{23}) + H_K(x_{23}) = \frac{c}{2 \text{sgn}(B(t))B(t)}((x_2 - x_2^*)^2 + (x_3 - x_3^*)^2). \quad (44)$$

According to the IDA-PBC technique [31], to reconstruct system (33) into a closed-loop system capable of asymptotically converging to equilibria, a mapping relationship is established

$$(J_d - R_d) \frac{\partial H_K}{\partial x_{23}}(x_{23}) + (J_a - R_a) \frac{\partial H}{\partial x_{23}}(x_{23}) = g_u \begin{bmatrix} 0 \\ u \end{bmatrix} + \Xi. \quad (45)$$

Substituting (32) and (34)–(43) into (45), the undetermined x_3^* is obtained by balancing the corresponding HRD channel of (45). This achieves virtual stabilization of the HRD channel in system (33). Further substituting (15a), (16), (21), and (27), x_3^* can be simplified as (19).

The u_R part is derived by substituting (19) and then balancing the corresponding RD1 channel of (45), that is

$$u_R = -\left(\frac{I}{B(t)} + \frac{rc}{\text{sgn}(B(t))B^2(t)}\right)(x_3 - x_3^*) - \left(\frac{1 + \text{sgn}(L)n(t)}{B(t)} + \frac{j(t)c}{\text{sgn}(B(t))B^2(t)}\right)(x_2 - x_2^*) - \frac{F(x_1, x_2, x_3, t)}{B(t)}. \quad (46)$$

At this point, by utilizing the virtual target x_3^* and controller u_R , (33) is stabilized as follows:

$$\dot{x}_{23} = [J_d - R_d] \frac{\partial H_d}{\partial x_{23}}(x_{23}) + g_u \begin{bmatrix} 0 \\ u_I \end{bmatrix}. \quad (47)$$

Stage 3: In this stage, the IA part u_I in (47) is designed to further enhance the performance of this approach.

Define an order-extended closed-loop PCH system associated to (47), with algebraic transformation between z and x_{23} to be determined [34]

$$\dot{z} = \left[\begin{array}{c|c} J_d - R_d & -K_I(t) \\ \hline K_I(t) & 0 \end{array} \right] \frac{\partial H_z}{\partial z}(z) \quad (48)$$

where $K_I(t)$ is the coefficient of IA, $z = [z_1, z_2, z_3]^T$, and the new Hamiltonian function is defined as

$$H_z(z) = H_d(z_1, z_2) + \frac{z_3^T K z_3}{2}, \quad (K > 0). \quad (49)$$

Set the algebraic transformation for z_1 and z_3

$$z_1 = x_2 \quad (50a)$$

$$z_3 = K_I(t) \int (x_2 - x_2^*) dt. \quad (50b)$$

Equation (50a) yields $\dot{z}_1 = \dot{x}_2$, indicating that the HRD channels of systems (47) and (48) are equivalent. By matching the

corresponding analytical expressions of these two channels, the algebraic transformation for z_2 is obtained as

$$z_2 = x_3 + \frac{K_I(t)K \operatorname{sgn}(B(t))B(t)}{(1 + \operatorname{sgn}(L)n(t)) \operatorname{sgn}(B(t))B(t) + j(t)c} z_3. \quad (51)$$

Take time derivative of both sides of (51), that is

$$\dot{z}_2 = \frac{\partial z_2}{\partial x_2} \dot{x}_2 + \frac{\partial z_2}{\partial x_3} \dot{x}_3 + \frac{\partial z_2}{\partial z_3} \dot{z}_3. \quad (52)$$

Substituting (50b), (51), the expressions of x_3 channel in (47) and z_2 channel in (48) into (52), the u_I part is derived as

$$u_I = -\frac{K_I(t)K \operatorname{sgn}(B(t))}{(1 + \operatorname{sgn}(L)n(t)) \operatorname{sgn}(B(t))B(t) + j(t)c} (x_2 - x_2^*) - \frac{(\operatorname{sgn}(B(t))B(t)I + rc)K_I(t)^2 K \operatorname{sgn}(B(t))}{(1 + \operatorname{sgn}(L)n(t))B^2(t) + \operatorname{sgn}(B(t))B(t)j(t)c} \int (x_2 - x_2^*) dt \\ ((1 + \operatorname{sgn}(L)n(t)) \operatorname{sgn}(B(t))B(t) + j(t)c \neq 0). \quad (53)$$

Finally refer to (37), the VO-IDA controller (18) is derived.

Stage 4: This stage proves that the closed-loop stability of (15) is guaranteed by utilizing the controller (18).

The whole analysis above clarifies that system (15) can be reconstructed to (48) through the utilizing of VO-IDA controller (18). Choose H_z in (49) as the Lyapunov function of system (48). Take time derivative of H_z along this system

$$\dot{H}_z = \frac{\partial^T H_z}{\partial z}(z) \left[\begin{array}{c|c} J_d - R_d & -K_I(t) \\ \hline K_I(t) & 0 \end{array} \right] \frac{\partial H_z}{\partial z}(z) \\ = -\frac{\operatorname{sgn}(L)n(t)c}{\operatorname{sgn}(B(t))B(t)L} (z_1 - z_1^*)^2 - \frac{(I + r)c}{\operatorname{sgn}(B(t))B(t)} (z_2 - z_2^*)^2 \\ < 0. \quad (54)$$

This indicates that system (48) achieves its closed-loop stability, i.e., z asymptotically stabilizes to the equilibria $(z_1^*, z_2^*, 0)$. Noting that the algebraic transformations between states x_{23} and z in (50a), (50b), and (51), the asymptotic stability of z is equivalent to x_{23} asymptotically stabilizing to (x_2^*, x_3^*) . Eventually, the virtual control relationship in (27) ensures that x_1 asymptotically converges to x_1^* . Thus, the proof of Theorem 1 is completed. ■

Remark 1: Equation (15) describes a high-order system with two HRD channels and without PCH structure. Given this, the VO-IDA approach employs backstepping and virtual stabilization techniques to reduce one HRD order of this system and stabilize it as a PCH structure (47) through introducing additional virtual relationships. This makes it possible to apply IDA-PBC and IA to system (15), avoiding six unknown PDEs to solve [35], while simultaneously enhancing control performance.

Remark 2: Equation (54) indicates that an autonomous system with a PCH structure is closed-loop stable. This is equivalent to the system power, characterized by the Lyapunov function, continuously dissipating and reaching the minimum at the target equilibria, which are, respectively, ensured by the properties of J_d , R_d in (12a) and the design of close-loop $H_d(x_{23})$, as satisfied in (47). Moreover, IA simultaneously retains the PCH structure and the same equilibria (x_2^*, x_3^*) of (47) for (48).

Noting that the internal stability upon CAV platoon control is a specific subproblem discussed in Theorem 1, Corollary 1 is presented as follows.

Corollary 1: Consider a CAV platoon with dynamic (1) and a reference trajectory (13) for the i th follower. Let $g_i = (1/\tau_i)$. If there exist parameters k_i , c_i , r_i , K_i , $K_{I,i}$, n_i and j_i , along with g_i , satisfying the following conditions:

$$k_i > 0, c_i > 0, r_i > 0, K_i > 0 \quad (55a)$$

$$n_i \in (0, 1) \cup (1, +\infty) \quad (55b)$$

$$(1 - n_i)g_i + j_i c_i \neq 0 \quad (55c)$$

then the motion states s_i , v_i , and a_i of the i th follower asymptotically stable to the equilibria (s_i^*, v_i^*, a_i^*) , by employing the PF communication topology, the CTH spacing policy (2), and the distributed VO-IDA controller u_i as follows:

$$u_i = u_{Ri} + u_{Ii} \quad (56)$$

where the IDA-PBC part u_{Ri} , IA part u_{Ii} , desired acceleration a_i^* and desired velocity v_i^* are given as

$$u_{Ri} = a_i^* - \frac{c_i r_i}{g_i^2} (a_i - a_i^*) + \left(\frac{n_i - 1}{g_i} - \frac{c_i j_i}{g_i^2} \right) (v_i - v_i^*) \quad (57)$$

$$u_{Ii} = -\frac{K_{I,i}^2 K_i}{(1 - n_i)g_i + j_i c_i} (v_i - v_i^*) - \frac{(g_i^2 + r_i c_i) K_{I,i}^2 K_i}{(1 - n_i)g_i^2 + j_i c_i g_i} \int (v_i - v_i^*) dt \quad (58)$$

$$a_i^* = \frac{n_i g_i (-k_i e_{1,i} + e_{2,i} + v_{i-1} - v_i^*) - c_i j_i h_i a_i}{(n_i - 1)g_i h_i - c_i j_i h_i} \quad (59)$$

$$v_i^* = s_i^* - k_i e_{1,i}. \quad (60)$$

IV. STRING STABILITY ANALYSIS AND ONLINE STABLE DOMAIN OPTIMIZATION

Since Section III only ensures internal stability, an explicit analysis of SFSS under heterogeneous dynamics is given in this section. This further motivates the combination of an online MOGA to guarantees the string stability properties for the VO-IDA controller under heterogeneous dynamics.

A. String Stability Analysis of Heterogeneous Dynamics

Theorem 2: Consider a CAV platoon with the heterogeneous dynamic (1) for the i th follower. Under the CTH spacing policy (2), the PF communication topology and the distributed VO-IDA controller (56), there are a set of conditions

$$(1 - n_i)g_i + j_i c_i \neq 0 \quad \wedge \quad m_{5i} > 0 \quad \wedge \quad m_{0i} \geq 0 \quad (61)$$

$$\text{case}_1 \quad \wedge \quad \rho \left(-\frac{m'_{4i}}{4m'_{5i}} \right) \geq 0 \quad (62)$$

$$\text{case}_2 \quad \wedge \quad \rho \left(\frac{-m'_{4i} M_{2i} + 9M_{1i}}{4m'_{5i} M_{2i}} \right) \geq 0 \quad \wedge \quad \rho \left(\frac{-m'_{4i} M_{2i} - 3M_{1i}}{4m'_{5i} M_{2i}} \right) \geq 0 \quad (63)$$

$$\text{case}_3 \quad \wedge \quad \rho \left(\frac{-m'_{4i} \pm \sqrt{M_{2i}}}{4m'_{5i}} \right) \geq 0 \quad (64)$$

$$\text{case}_4 \wedge \rho \left(\frac{-m'_{4i} + \frac{2M_{1i}M_{5i}}{M_{4i}} \pm \sqrt{\frac{2M_{4i}}{M_{5i}}}}{4m'_{5i}} \right) \geq 0 \wedge$$

$$\rho \left(\frac{-m'_{4i} - \frac{2M_{1i}M_{5i}}{M_{4i}}}{4m'_{5i}} \right) \geq 0 \quad (65)$$

$$\text{case}_5 \wedge \rho \left(\frac{-m'_{4i} - \frac{2M_{1i}M_{5i}}{M_{4i}}}{4m'_{5i}} \right) \geq 0 \quad (66)$$

$$\text{case}_6 \wedge \rho \left(\frac{1}{4m'_{5i}} \left(\text{sgn}(M_{1i}) \sqrt{\frac{M_{2i} + \sqrt[3]{W_{1i}} + \sqrt[3]{W_{2i}}}{3}} \right. \right. \\ \left. \left. - m'_{4i} \pm \sqrt{\frac{2M_{2i} - (\sqrt[3]{W_{1i}} + \sqrt[3]{W_{2i}}) + 2\sqrt{W_i}}{3}} \right) \right) \geq 0 \quad (67)$$

$$\text{case}_7 \wedge \rho \left(\frac{-m'_{4i} \pm \sqrt{M_{2i} + 2\sqrt{M_{0i}}}}{4m'_{5i}} \right) \geq 0 \wedge \\ \rho \left(\frac{-m'_{4i} \pm \sqrt{M_{2i} - 2\sqrt{M_{0i}}}}{4m'_{5i}} \right) \geq 0 \quad (68)$$

$$\text{case}_8 \wedge \rho \left(\frac{-m'_{4i} + \text{sgn}(M_{1i})\sqrt{y_{1i}} \pm (\sqrt{y_{2i}} + \sqrt{y_{3i}})}{4m'_{5i}} \right) \geq 0 \\ \wedge \rho \left(\frac{-m'_{4i} - \text{sgn}(M_{1i})\sqrt{y_{1i}} \pm (\sqrt{y_{2i}} - \sqrt{y_{3i}})}{4m'_{5i}} \right) \geq 0 \quad (69)$$

$$\begin{cases} 1, & \sum_{i'=1}^8 \text{case}_{i'} = 0 \\ 0, & \sum_{i'=1}^8 \text{case}_{i'} \neq 0 \end{cases} \quad (70)$$

where

$\text{cond}_1 \wedge \text{cond}_2$ denotes an ‘and’ logic

$$\text{case}_1 = (M_{0i} = M_{1i} = M_{2i} = 0) \quad (71)$$

$$\text{case}_2 = (M_{0i}M_{1i}M_{2i} \neq 0 \wedge M_{3i} = M_{4i} = M_{5i} = 0) \quad (72)$$

$$\text{case}_3 = (M_{0i} = M_{1i} = 0 \wedge M_{2i} > 0) \quad (73)$$

$$\text{case}_4 = (M_{3i}M_{4i}M_{5i} \neq 0 \wedge \Delta_i = 0 \wedge M_{4i}M_{5i} > 0) \quad (74)$$

$$\text{case}_5 = (M_{3i}M_{4i}M_{5i} \neq 0 \wedge \Delta_i = 0 \wedge M_{4i}M_{5i} < 0) \quad (75)$$

$$\text{case}_6 = (\Delta_i > 0) \quad (76)$$

$$\text{case}_7 = (\Delta_i < 0 \wedge M_{0i} > 0 \wedge M_{1i} = 0 \wedge M_{2i} > 0) \quad (77)$$

$$\text{case}_8 = (\Delta_i < 0 \wedge M_{0i} > 0 \wedge M_{1i} \neq 0 \wedge M_{2i} > 0) \quad (78)$$

$$\rho(x) = f(x \cdot \epsilon(x)), \quad \epsilon(x) \text{ is a heaviside function} \quad (79)$$

$$f(x) = \sum_{\xi=0}^5 m_{\xi i} x^{\xi} \quad (80)$$

$$W_{1i,2i} = M_{2i}M_{5i} + 3 \left(\frac{-M_{4i} \pm \sqrt{\Delta_i}}{2} \right) \quad (81)$$

$$W_i = M_{2i}^2 - M_{2i} \left(\sqrt[3]{W_{1i}} + \sqrt[3]{W_{2i}} \right) + \left(\sqrt[3]{W_{1i}} + \sqrt[3]{W_{2i}} \right)^2 \\ - 3M_{5i} \quad (82)$$

$$\theta_i = \arccos \frac{3M_{4i} - 2M_{2i}M_{5i}}{2M_{5i}\sqrt{M_{5i}}} \quad (83)$$

$$y_{1i} = \frac{M_{2i} - 2\sqrt{M_{5i}} \cos \frac{\theta_i}{3}}{3} \quad (84)$$

$$y_{2i,3i} = \frac{M_{2i} - 2\sqrt{M_{5i}} \cos \left(\frac{\theta_i}{3} \pm \frac{2\pi}{3} \right)}{3} \quad (85)$$

$$M_{0i} = 3m_{4i}^4 + 16m_{5i}^2m_{3i}^2 - 16m_{5i}^2m_{4i}^2m_{3i} + 16m_{5i}^2m_{4i}^2m_{2i} \\ - 64m_{5i}^3m_{1i} \quad (86)$$

$$M_{1i} = -m_{4i}^3 + 4m_{5i}^2m_{4i}^2m_{3i} - 8m_{5i}^2m_{2i}^2 \quad (87)$$

$$M_{2i} = 3m_{4i}^2 - 8m_{5i}^2m_{3i} \quad (88)$$

$$M_{3i} = M_{0i}^2 - 3M_{2i}M_{1i}^2 \quad (89)$$

$$M_{4i} = M_{2i}M_{0i} - 9M_{1i}^2 \quad (90)$$

$$M_{5i} = M_{2i}^2 - 3M_{0i} \quad (91)$$

$$\Delta_i = M_{4i}^2 - 4M_{5i}M_{3i} \quad (92)$$

$$m'_{\xi'i} = \xi' m_{\xi'i}, \quad (\xi' = 1, 2, \dots, 5) \quad (93)$$

$$m_{0i} = \varphi_{i,0}^{\text{own}2} - \varphi_{i-1,0}^{\text{pre}2} \quad (94)$$

$$m_{1i} = 2\varphi_{i,0}^{\text{own}}\varphi_{i,2}^{\text{own}} + 2\varphi_{i-1,0}^{\text{pre}}\varphi_{i-1,2}^{\text{pre}} + \varphi_{i,1}^{\text{own}2} - \varphi_{i-1,1}^{\text{pre}2} \quad (95)$$

$$m_{2i} = 2\varphi_{i,0}^{\text{own}}\varphi_{i,4}^{\text{own}} + 2\varphi_{i,1}^{\text{own}}\varphi_{i,3}^{\text{own}} + 2\varphi_{i-1,1}^{\text{pre}}\varphi_{i-1,3}^{\text{pre}} + \varphi_{i,2}^{\text{own}2} \\ - \varphi_{i-1,2}^{\text{pre}2} \quad (96)$$

$$m_{3i} = 2\varphi_{i,2}^{\text{own}}\varphi_{i,4}^{\text{own}} + 2\varphi_{i,1}^{\text{own}}\varphi_{i,5}^{\text{own}} + \varphi_{i,3}^{\text{own}2} - \varphi_{i-1,3}^{\text{pre}2} \quad (97)$$

$$m_{4i} = \varphi_{i,4}^{\text{own}2} + 2\varphi_{i,3}^{\text{own}}\varphi_{i,5}^{\text{own}} \quad (98)$$

$$m_{5i} = \varphi_{i,5}^{\text{own}2} \quad (99)$$

$$\varphi_{i-1}^{\text{ref}} = \begin{bmatrix} k_{i-1}P_{1,i-1}P_{3,i} \\ k_{i-1}P_{2,i-1}P_{3,i} + P_{1,i-1}P_{3,i} + k_{i-1}P_{1,i-1} \\ P_{2,i-1}P_{3,i} + k_{i-1}P_{2,i-1} + P_{1,i-1} \\ P_{2,i-1} \end{bmatrix} \quad (100)$$

$$\varphi_i^{\text{own}} = \begin{bmatrix} k_iP_{1,i}P_{3,i-1} \\ h_i k_i P_{1,i}P_{3,i-1} + k_i P_{2,i}P_{3,i-1} + P_{1,i}P_{3,i-1} \\ + k_i P_{1,i} \\ h_i k_i P_{2,i}P_{3,i-1} + h_i P_{1,i}P_{3,i-1} + P_{2,i}P_{3,i-1} \\ + h_i k_i P_{1,i} + P_{2,i}k_i + P_{1,i} \\ h_i P_{2,i}P_{3,i-1} + P_{3,i-1}P_{3,i} + h_i k_i P_{2,i} \\ + h_i P_{1,i} + P_{2,i} \\ h_i P_{2,i} + P_{3,i-1} + P_{3,i} \\ 1 \end{bmatrix} \quad (101)$$

$$P_{1,i} = \frac{(g_i^2 + r_i c_i) K_{L,i}^2 K_i}{(1 - n_i)g_i + j_i c_i} \quad (102)$$

$$P_{2,i} = \frac{(g_i^2 + r_i c_i)n_i + g_i h_i K_{L,i}^2 K_i}{(1 - n_i)g_i h_i + j_i c_i h_i} + \frac{(1 - n_i)g_i + j_i c_i}{g_i} \quad (103)$$

$$P_{3,i} = \frac{g_i^2 + c_i r_i}{(1 - n_i)g_i + j_i c_i} \quad (104)$$

The attenuation of SFSS (5) holds if and only if (61) is true, and any of the conditions (62)–(70) are true.

Proof: Combining (13), (22), and (27), analogous to (28) in Theorem 1, it yields

$$v_i + h_i a_i - v_{i-1} = e_{2,i} - k_i e_{1,i}. \quad (105)$$

Substituting (56)–(58) into (1c), and then using (105) and (59) for simplification, we obtain

$$P_{1,i} \int (v_i - v_i^*) dt + P_{2,i}(v_i - v_i^*) + P_{3,i}a_i + \dot{a}_i = 0 \quad (106)$$

where $j_i c_i + (1 - n_i)g_i \neq 0$ in (61) avoids the singularity for $P_{1,i}$, $P_{2,i}$, and $P_{3,i}$. By substitute (60) and then take derivative, (106) becomes

$$P_{2,i}\ddot{e}_{1,i} + (P_{1,i} + k_i P_{2,i})\dot{e}_{1,i} + P_{1,i}k_i e_{1,i} + \ddot{a}_i + P_{3,i}\dot{a}_i = 0. \quad (107)$$

Similarly calculations of (106) and (107) for the predecessor $P_{2,i-1}\ddot{e}_{1,i-1} + (P_{1,i-1} + k_{i-1}P_{2,i-1})\dot{e}_{1,i-1} + P_{1,i-1}k_{i-1}e_{1,i-1} + \ddot{a}_{i-1} + P_{3,i}\ddot{a}_{i-1} + (P_{3,i-1} - P_{3,i})\dot{a}_{i-1} = 0$. (108)

Take derivative of (107)

$$P_{2,i}e_{1,i}^{(3)} + (P_{1,i} + k_i P_{2,i})\ddot{e}_{1,i} + P_{1,i}k_i\dot{e}_{1,i} + a_i^{(3)} + P_{3,i}\ddot{a}_i = 0. \quad (109)$$

Calculate (107)–(108) + $h \times$ (109)

$$\begin{aligned} & e_{1,i}^{(4)} + (h_i P_{2,i} + P_{3,i})e_{1,i}^{(3)} + (h_i P_{1,i} + P_{2,i} + h_i k_i P_{2,i})\ddot{e}_{1,i} \\ & + (P_{1,i} + h_i k_i P_{1,i} + k_i P_{2,i})\dot{e}_{1,i} + k_i P_{1,i}e_{1,i} \\ & = P_{2,i-1}\ddot{e}_{1,i-1} + (P_{1,i-1} + k_{i-1}P_{2,i-1})\dot{e}_{1,i-1} \\ & + k_{i-1}P_{1,i-1}e_{1,i-1} + (P_{3,i-1} - P_{3,i})\dot{a}_{i-1}. \end{aligned} \quad (110)$$

Take Laplace transformation for (108) and (110), then substituting $\dot{a}_{i-1}$ by using former into latter, that yields

$$\theta_{i-1}^{\text{ref}}(s)\varphi_{i-1}^{\text{ref}}e_{1,i-1}(s) = \theta_i^{\text{own}}(s)\varphi_i^{\text{own}}e_{1,i}(s) \quad (111)$$

where $\theta_{i-1}^{\text{ref}}(s) = [1, s, s^2, s^3]$, $\theta_i^{\text{own}}(s) = [1, s, s^2, s^3, s^4, s^5]$, $\varphi_{i-1}^{\text{ref}}$, and φ_i^{own} are referenced to (100) and (101).

In view of string stability, the strict $\mathcal{L}_2$ string stability upon frequency domain [43] is employed as

$$\|e_{1,i}(t)\|_{\mathcal{L}_2} \leq \|e_{1,i-1}(t)\|_{\mathcal{L}_2}. \quad (112)$$

Then analyze the behavior of input $e_{1,i}$ and output $e_{1,i-1}$ for the i th follower, based on (111) and Definition 2, define

$$G_{i-1,i}(s) = \frac{e_{1,i}(s)}{e_{1,i-1}(s)} = \frac{\theta_{i-1}^{\text{ref}}(s)\varphi_{i-1}^{\text{ref}}}{\theta_i^{\text{own}}(s)\varphi_i^{\text{own}}}. \quad (113)$$

According to Lemma 1, one gets

$$\|G_{i-1,i}(s)\|_{\mathcal{H}_\infty} = \|G_{i-1,i}(j\omega)\|_{\mathcal{H}_\infty}. \quad (114)$$

By employing Parseval theorem [7] and (113), one gets

$$\begin{aligned} \|e_{1,i}(t)\|_{\mathcal{L}_2}^2 &= \frac{1}{2\pi} \int_0^{+\infty} |e_{1,i}(j\omega)|^2 d\omega \\ &= \frac{1}{2\pi} \int_0^{+\infty} |G_{i-1,i}(j\omega)e_{1,i-1}(j\omega)|^2 d\omega \end{aligned} \quad (115)$$

further by introducing Cauchy–Schwarz inequality, we obtain

$$\begin{aligned} \|e_{1,i}(t)\|_{\mathcal{L}_2}^2 &= \frac{1}{2\pi} \int_0^{+\infty} |(G_{i-1,i}(j\omega)e_{1,i-1}(j\omega), 1)|^2 d\omega \\ &\leq \frac{1}{2\pi} \int_0^{+\infty} (e_{1,i-1}^H(j\omega)G_{i-1,i}^H(j\omega)G_{i-1,i}(j\omega)e_{1,i-1}(j\omega))d\omega \\ &\leq \|G_{i-1,i}(j\omega)\|_{\mathcal{H}_\infty}^2 \|e_{1,i-1}(t)\|_{\mathcal{L}_2}^2 \end{aligned} \quad (116)$$

where $(\cdot, \cdot)$ denotes inner product and H denotes the conjugate transpose of a vector.

Analogize to (9), equation (116) is not an equality anymore. This is because the presence of the first-order inertial time lag τ_i , which reduces the $\mathcal{L}_2$ norm of the output term compared to a scenario without this lag.

As a result, (5) serves as a sufficient condition for the imposed requirement (112) under the control problem of vehicle platoon with linear dynamics and PF communicating topology, though some degree of conservatism is introduced.

The work above provides a basis for later explicitly analysis of string stability under heterogeneous dynamics.

Substitute (100) and (101) into (113) and combine (114), the $\mathcal{H}_\infty$ norm of $G_{i-1,i}(j\omega)$ is calculated as

$$\|G_{i-1,i}(j\omega)\|_{\mathcal{H}_\infty}^2 = \sup_{\omega>0} \frac{\|\theta_{i-1}^{\text{ref}}(j\omega)\varphi_{i-1}^{\text{ref}}\|_{\mathcal{L}_2}^2}{\|\theta_i^{\text{own}}(j\omega)\varphi_i^{\text{own}}\|_{\mathcal{L}_2}^2} = \sup_{\omega>0} \frac{D_{i-1}^{\text{ref}}}{D_i^{\text{own}}} \leq 1 \quad (117)$$

where

$$\begin{aligned} D_{i-1}^{\text{ref}} &= \varphi_{i-1,3}^{\text{ref}}{}^2 \omega^6 + \left(\varphi_{i-1,2}^{\text{ref}}{}^2 - 2\varphi_{i-1,1}^{\text{ref}}\varphi_{i-1,3}^{\text{ref}} \right) \omega^4 \\ &+ \left(\varphi_{i-1,1}^{\text{ref}}{}^2 - 2\varphi_{i-1,0}^{\text{ref}}\varphi_{i-1,2}^{\text{ref}} \right) \omega^2 + \varphi_{i-1,0}^{\text{ref}}{}^2 \\ D_i^{\text{own}} &= \varphi_{i,5}^{\text{own}}{}^2 \omega^{10} + \left(\varphi_{i,4}^{\text{own}}{}^2 - 2\varphi_{i,3}^{\text{own}}\varphi_{i,5}^{\text{own}} \right) \omega^8 \\ &+ \left(\varphi_{i,3}^{\text{own}}{}^2 - 2\varphi_{i,2}^{\text{own}}\varphi_{i,4}^{\text{own}} + 2\varphi_{i,1}^{\text{own}}\varphi_{i,5}^{\text{own}} \right) \omega^6 \\ &+ \left(\varphi_{i,2}^{\text{own}}{}^2 - 2\varphi_{i,1}^{\text{own}}\varphi_{i,3}^{\text{own}} + 2\varphi_{i,0}^{\text{own}}\varphi_{i,4}^{\text{own}} \right) \omega^4 \\ &+ \left(\varphi_{i,1}^{\text{own}}{}^2 - 2\varphi_{i,0}^{\text{own}}\varphi_{i,2}^{\text{own}} \right) \omega^2 + \varphi_{i,0}^{\text{own}}{}^2. \end{aligned} \quad (118) \quad (119)$$

Then (117) holds if and only if the function in (80) satisfies

$$f(\omega^2) = D_{i-1}^{\text{ref}} - D_i^{\text{own}} = \sum_{\xi=0}^5 m_{\xi i} \omega^{2\xi} \geq 0 \quad \forall \omega^2 \in (0, +\infty) \quad (120)$$

where $m_{\xi i}$ ($\xi = 0, 1, \dots, 5$) refer to (94)–(99).

The derivative of (120) is calculated as

$$f'(\omega^2) = \sum_{\xi'=1}^5 m'_{\xi' i} \omega^{2(\xi'-1)} \geq 0 \quad \forall \omega^2 \in (0, +\infty) \quad (121)$$

where $m'_{\xi' i}$ ($\xi' = 1, 2, \dots, 5$) refer to (93).

The $m_{5i} > 0$ in (61) determines that $f(\omega^2)$ is constantly positive on $\omega^2 \in [U, +\infty)$ (where U is a positive value that inevitably exist). On this basis, (120) holds if and only if the nonnegativity properties for intercept of $f(\omega^2)$ and the extreme points of $f(\omega^2)$ in positive semi-axis are guaranteed simultaneously, with the former of which ensured by $m_{0i} \geq 0$.

The extreme points of $f(\omega^2)$ are equivalent to the real solutions of $f'(\omega^2)$, and the distribution of latter can be derived by criterions in Tianheng formula [44]. Meanwhile, the ρ function in (79) is designed to transfer the solutions of $f'(\omega^2)$ distributed in negative semi-axis to the original point of $f(\omega^2)$. Conditions (62)–(78) are eventually obtained through synthesizing the ρ function and distribution of those real solutions, with corresponding auxiliary variables (81)–(92). Thus, the proof of Theorem 2 is completed. ■

Note Theorem 2 considers the heterogeneous dynamics, whereas the homogeneous assumption for dynamics and controller parameters is a simpler case with all the follower vehicles ($i = 1 \dots N$) have the same τ_i , $P_{1,i}$, $P_{2,i}$, $P_{3,i}$, $\varphi_{i-1}^{\text{ref}}$, φ_i^{own} , $m_{\xi i}$ ($\xi = 0, 1, \dots, 5$), $m'_{\xi' i}$ ($\xi' = 1, 2, \dots, 5$). The $m_{0i} = 0$ implies that $f(x)$ in (80) is degraded into a quadratic function. Combined with Shengjin formula [45], and under the homogenous assumption, Corollary 2 is given as follows.

Corollary 2: Consider a CAV platoon with the homogenous dynamic (1) for the i th follower. Under the CTH spacing policy (2), the PF communication topology and the distributed VO-IDA controller (56), there are a set of conditions

$$j_i c_i + (1 - n_i) g_i \neq 0 \quad \wedge \quad m_{5i} > 0 \quad \wedge \quad m_{1i} \geq 0 \quad (122)$$

$$\text{case}_1 \quad \wedge \quad \rho \left(\frac{-3m'_{2i}}{m'_{3i}} \right) \geq 0 \quad (123)$$

$$\text{case}_2 \quad \wedge \quad \rho \left(\frac{-m'_{4i} - \sqrt[3]{Y_1} - \sqrt[3]{Y_2}}{3m'_{5i}} \right) \geq 0 \quad (124)$$

$$\text{case}_3 \quad \wedge \quad \rho \left(\frac{-m'_{4i}}{m'_{5i}} + K'_i \right) \geq 0 \quad \wedge \quad \rho \left(-\frac{1}{2} K_i \right) \geq 0 \quad (125)$$

$$\text{case}_4 \quad \wedge \quad \rho \left(\frac{-m'_{4i} - 2\sqrt{M_{2i}} \cos \frac{\theta_i}{3}}{3m'_{5i}} \right) \geq 0 \quad \wedge \quad \rho \left(\frac{-m'_{4i} + \sqrt{M_{2i}} \left(\cos \frac{\theta_i}{3} \pm \sqrt{3} \sin \frac{\theta_i}{3} \right)}{3m'_{5i}} \right) \geq 0 \quad (126)$$

where

$$\text{case}_1 = (M_{1i} = M_{2i} = 0)$$

$$\text{case}_2 = (\Delta_i = M_{1i}^2 - 4M_{2i}M_{0i} > 0)$$

$$\text{case}_3 = (\Delta_i = M_{1i}^2 - 4M_{2i}M_{0i} = 0)$$

$$\text{case}_4 = (\Delta_i = M_{1i}^2 - 4M_{2i}M_{0i} < 0)$$

$$\rho(x) \text{ is in accord with (79)}$$

$$Y_{1,2} = M_{2i}m'_{4i} + 3m'_{5i} \left(\frac{-M_{1i} \pm \sqrt{M_{1i}^2 - 4M_{2i}M_{0i}}}{2} \right)$$

$$K'_i = \frac{M_{1i}}{M_{2i}} (M_{2i} \neq 0)$$

$$T_i = \frac{2M_{2i}m'_{4i} - 3m'_{5i}M_{1i}}{2\sqrt{M_{2i}^3}}, \quad (M_{2i} > 0, -1 < T_i < 1)$$

$$\theta_i = \arccos T_i$$

$$M_{0i} = m_{3i}^2 - 3m'_{4i}m'_{2i}$$

$$M_{1i} = m'_{4i}m'_{3i} - 9m_{3i}^1m'_{2i}$$

$$M_{2i} = M_{1i}^2 - 3m'_{3i}m'_{3i}$$

$$\Delta_i = M_{1i}^2 - 4M_{2i}M_{0i}$$

$$m'_{\xi i} = \xi m_{\xi i} \quad (\xi = 2, 3, \dots, 5)$$

$$m_{\xi i}, \varphi_{i-1}^{\text{ref}}, \varphi_i^{\text{own}}, P_{1,i}, P_{2,i}, P_{3,i}, (\xi = 1, 2, \dots, 5) \text{ are in accord with (95)-(104).}$$

The attenuation of SFSS (5) holds if and only if (122) is true, and any of the conditions (123)–(126) are true.

To guarantee the internal stability (14) and the attenuation property of string stability (5) simultaneously, certain key control parameters shall be specified and valued within a stable domain. Please note, the internal stability is inherently guaranteed with the VO-IDA controller (56) and condition (55) implemented, as referenced to Theorem 1. Consequently, the

corresponding stable domain of each vehicle under homogeneous/heterogeneous dynamics is solely determined by Corollary 2 and Theorem 2, respectively.

Remark 3: Consider the following nonlinear system, which is also commonly used for the vehicle platoons [23]:

$$\dot{s}_i = v_i \quad (127a)$$

$$\dot{v}_i = a_i \quad (127b)$$

$$\dot{a}_i = -\frac{1}{\tau_i} a_i + \frac{1}{\tau_i m_i} \left(\frac{\eta_{T,i}}{R_i} u_i - C_{A,i} v_i (2\tau_i \dot{v}_i + v_i) - m_i g (f_i \cos \alpha_i + \sin \alpha_i) \right) \quad (127c)$$

where u_i here denotes the desired driving/braking torque, m_i is the mass, $C_{A,i}$ is the lumped aerodynamic drag coefficient, g is the gravitational acceleration, f_i is the rolling resistance coefficient, T_i is the actual driving/braking torque, τ_i is the lumped time lag, R_i is the tire radius, $\eta_{T,i}$ is the mechanical efficiency of the powertrain, and α_i is the road slope. The VO-IDA is also applicable to nonlinear system (127). By following the parameters and conditions (55a)–(55c) from Corollary 1, one can arrive at the conclusions as expressed in (57)–(60), where (57) and (58) are replaced by

$$u_{Ri} = \frac{R_i}{\eta_{T,i}} (C_{A,i} v_i (2\tau_i \dot{v}_i + v_i) + m_i g (f_i \cos \alpha_i + \sin \alpha_i)) + \frac{m_i R_i}{\eta_{T,i}} a_i^* - \frac{m_i R_i c_i r_i}{\eta_{T,i} g_i^2} (a_i - a_i^*) + \frac{m_i R_i}{\eta_{T,i}} \left(\frac{n_i - 1}{g_i} - \frac{c_i j_i}{g_i^2} \right) (v_i - v_i^*) \quad (128)$$

$$u_{Li} = -\frac{K_{L,i}^2 K_i m_i R_i}{(1 - n_i) g_i \eta_{T,i} + j_i c_i \eta_{T,i}} (v_i - v_i^*) - \frac{(g_i^2 + r_i c_i) K_{L,i}^2 K_i m_i R_i}{(1 - n_i) g_i^2 \eta_{T,i} + j_i c_i g_i \eta_{T,i}} \int (v_i - v_i^*) dt. \quad (129)$$

The analysis and conclusion of SFSS remain unchanged.

B. Strategy for Online Stable Domain Optimization

This section elaborates a guideline for VO-IDA controllers to online self-learn their appropriate target locations within the corresponding stable domains, in pursuit of nice string stability properties under heterogeneous dynamics.

Equation (107) is further expressed as (130) through Laplace transform, and similarly with the predecessor

$$a_i(s) = -\frac{P_{2,i} \ddot{e}_{1,i}(s) + (P_{1,i} + k_i P_{2,i}) \dot{e}_{1,i}(s) + k_i P_{1,i} e_{1,i}(s)}{s^2 + P_{3,i} s}. \quad (130)$$

Equation (130) and the predecessor one are made quotient as

$$\frac{a_i(s)}{a_{i-1}(s)} = \frac{\hat{\varphi}_i^{\text{own}}[s^2, s, 1]^T s + P_{3,i-1}}{\hat{\varphi}_{i-1}^{\text{ref}}[s^2, s, 1]^T s + P_{3,i}} \frac{e_{1,i}(s)}{e_{1,i-1}(s)} \quad (131)$$

where

$$\hat{\varphi}_{i-1}^{\text{ref}} = [P_{2,i-1}, P_{1,i-1} + k_{i-1} P_{2,i-1}, k_{i-1} P_{1,i-1}]$$

$$\hat{\varphi}_i^{\text{own}} = [P_{2,i}, P_{1,i} + k_i P_{2,i}, k_i P_{1,i}].$$

Considering the homogeneous assumption in Corollary 2, (131) is simplified as

$$\frac{a_i(s)}{a_{i-1}(s)} = \frac{e_{1,i}(s)}{e_{1,i-1}(s)}. \quad (132)$$

Equation (132) reveals the consistency property (6) for platoons with homogeneous dynamics and controller parameters. The attenuation and consistency properties remain equivalent with regard to spacing errors and accelerations. Indeed, the attenuation property has been ensured, as seen in (116), where $\|G_{i-1,i}(j\omega)\|_{H_\infty}^2 = 1$, whereas interaccelerations are led to close trend and levels by the guaranteed internal stability, but contradicts against the attenuation property (5).

In contrast to (132), heterogeneous dynamics are considered in (131). Noticing that $\hat{\phi}_{i(i-1)}^{\text{own(pre)}}$ consists of $P_{1,i(i-1)}$ and $P_{2,i(i-1)}$ in a linear relationship, a set of objectives are set as

$$P_{1,i} \rightarrow P_{1,i}^* = \frac{1}{\eta_i} P_{1,i-1}, P_{2,i} \rightarrow P_{2,i}^* = \frac{1}{\eta_i} P_{2,i-1} \quad (133)$$

$$P_{3,i} \rightarrow P_{3,i}^* = P_{3,i-1} \quad (134)$$

where $\eta_i \in (0, 1]$ is attenuation factor, which can be manually modulated to enhance attenuation property. By substituting (133) and (134), then (131) becomes

$$\frac{a_i(s)}{a_{i-1}(s)} = \frac{e_{1,i}(s)}{\eta_i e_{1,i-1}(s)}. \quad (135)$$

Equation (135) reveals the consistency property (6) for platoons with heterogeneous dynamics and controller parameters. Assume that $P_{i,1}$, $P_{i,2}$ and $P_{3,i}$ reach their respective target as (133) and (134) under heterogeneous dynamics, simultaneously. When $\eta_i = 1$, it is judged that the spacing error always maintains relationship (132), just as in the homogeneous case. When $\eta_i < 1$, this relationship further becomes (135), revealing a stronger attenuation intensity for spacing error compared to (132). However, $P_{i,1}$, $P_{i,2}$ and $P_{3,i}$ are all affected by heterogeneous τ_i , which can be eliminated by proper configurations of controller parameters. This configuration requires an exploration for the target position within the corresponding stable domain. Select the key control parameters n_i , j_i , and $K_{I,i}$, which are derived from stage 1, stage 2, and stage 3 in Theorem 1, respectively.

The complexity of formulations in (61)–(70) makes it difficult to characterize the stable domain with exact analytic formulas. In order to fulfil the exploration demand, a modified MOGA is introduced here to online self-learn where the target position locates. The procedure is illustrated in Algorithm 1. Since the first follower vehicle are only able to use the motion information from leader vehicle, the key controller parameters of which are required to be valued manually by Corollary 2.

Compared to the regular MOGA algorithm [46], [47], a couple of notes on the modifications are given here.

- 1) The function $Obj_{\xi,i}$ is designed with reference to (133) and (134).
- 2) The stable domain determined by Theorem 2 is converted to a single logical criterion $logi_{in,i}$, which is introduced as a penalty term into the fitness function to force the population to cluster within the stability domain.

Algorithm 1: Modified MOGA

Input: Key controller parameters n_{i-1} , j_{i-1} , $K_{I,i-1}$

Others η_i , c_i , r_i , k_i , K_i , h_{i-1} , h_i , τ_{i-1} , τ_i

Output: $n_i \in \Theta_{n_i}$, $j_i \in \Theta_{j_i}$, $K_{I,i} \in \Theta_{K_{I,i}}$

```

1 Initialize the population
2 for  $q = 2$  to  $G$  do
    Constrained Fitness Design
3     Compute objectives as (133) and (134):
        $Obj_{\xi,i} = -|P_{\xi,i} - P_{\xi,i}^*| + Upper_{\xi,i}$  ( $\xi = 1, 2, 3$ )
4     Convert the population lying within stable domain of
       Theorem 2 or not to logical values  $logi_{in,i}$ 
5     Translate fitness:  $Fit_{\xi,i} = Obj_{\xi,i} - \lambda_{\xi,i} logi_{in,i}$ 
6     Selection, then crossover and mutation
7     Repeat lines 3-5 to generate transitional population
    Updating of Pareto pool  $\Upsilon$ 
8     for  $q' = 1$  to  $n_{pop}$  do
9         if  $\prod_{j=1}^{\xi} \text{isempty}(\text{intersectn}(index_{\xi,i}))$  then
10             Introduce the new Pareto solutions into  $\Upsilon$ 
11     Clear the highly approximate solutions in  $\Upsilon$ 
12     Exclude the dominated solutions from  $\Upsilon$ 
13     Renew the population through elitism
14     Select the best  $n_i$ ,  $j_i$ ,  $K_{I,i}$  from final  $\Upsilon$  that
       maximizes  $Fit_{\xi,i}$ 
15     while  $Fit_{\xi,i}$  converges during  $\delta_q$  iterations do
16         break

```

- 3) The “ $index_{\xi,i}$ ” records the index of individuals among population that are nondominant with respect to $Fit_{\xi,i}$. The “intersectn” in line 9 is designed for the j th objective to search better solutions among population by fixing the other $\xi - 1$ objectives, which fullfills the updating of pareto pool. Lines 11 and 12 work on a similar principle.

Remark 4: Owing to the virtual stabilization in Theorem 1, the desired acceleration is reformulated in a manner such that the entire VO-IDA controller becomes independent of the a_{i-1} . This independence, further combined with the feedback attribute of IDA-PBC, allows the acceleration and spacing error to satisfy (131). This is a pivotal aspect for the controller to equip with optimization algorithms like MOGA.

Up to this point, the entire workflow of this research is summarized in Fig. 1.

V. SIMULATION RESULTS

In order to verify the superiority of proposed approach on the tracking accuracy, the convergence, the ability to handle with heterogenous dynamics, as well as the attenuation and consistency upon string stability, comparative experiments are conducted in MATLAB/Simulink. The effectiveness of proposed approach is validated through Section V. Sections V-A–V-C, following a progressive sequence. Finally, an existing approach is taken into comparison in another Section V-D.

A platoon of CAVs consisting of one leader and seven follower vehicles with homogeneous/heterogeneous dynamics is considered, as shown in Fig. 1.

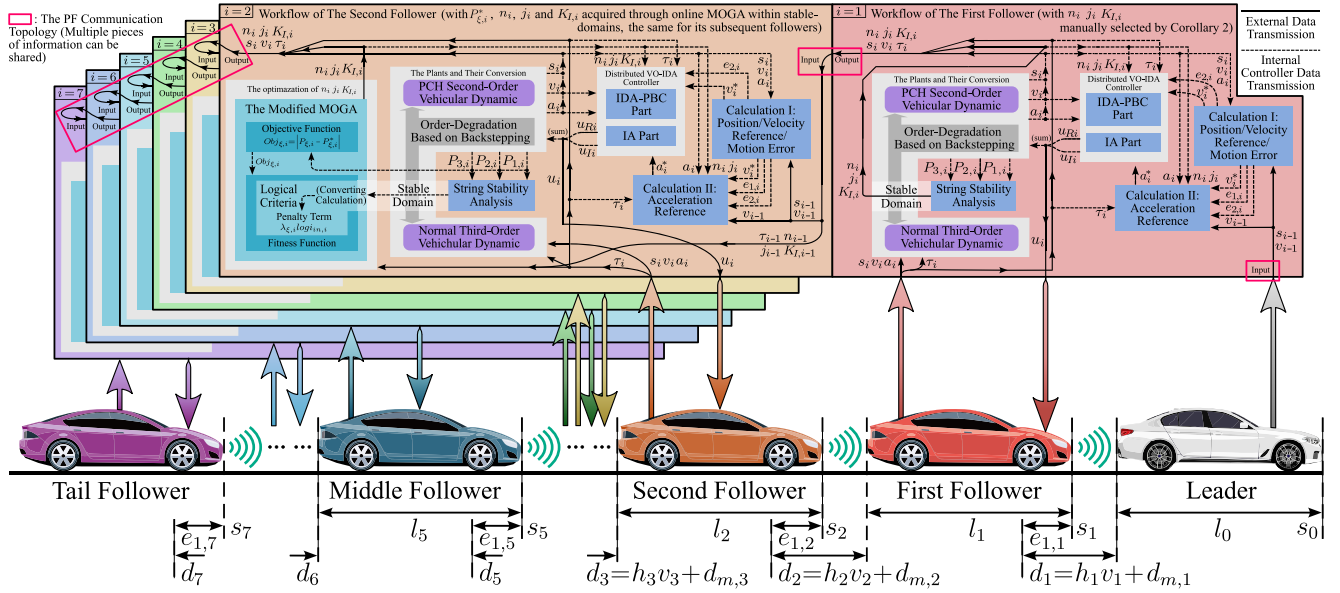


Fig. 1. Workflow of PCH-based approach for heterogeneous CAV platoons.

TABLE I
CONTROLLER PARAMETERS

Controller Parameter	Setting value
Backstepping gain k_i	3
Auxiliary variable n_i	5
IDA-PBC parameter j_i	8
IDA-PBC parameter r_i	1
Output vector parameter c_i	5
IA parameter K_i	5
IA parameter $K_{I,i}$	5

TABLE II
HOMOGENEOUS DYNAMICAL PARAMETERS

Parameter	Setting value
Time headway h_i	1.5s
Lumped time lag τ_i	0.2s
Minimum safety distance $d_{m,i}$	5m

TABLE III
HETEROGENEOUS DYNAMICAL PARAMETERS

Index i	$h_i(s)$	$\tau_i(s)$	$d_{m,i}(m)$
1	1.50	0.27	5
2	1.00	0.63	5
3	0.75	0.20	5
4	1.20	0.25	5
5	0.75	0.20	5
6	1.50	0.33	5
7	1.00	0.20	5

The initial positions and motion states of all the vehicles are $s_i = -5i + 35$ m, $v_i = 0$ m/s, $a_i = 0$ m/s² ($i = 0, 1, 2, \dots, 7$). The leader vehicle undergoes three stages of acceleration and deceleration. At 10 s, it accelerates from 0 to 20 m/s. At 53 s, it further accelerates to 90 m/s. From 61 to 74 s, its acceleration stabilizes at 2 m/s². At 117 s, an emergency braking occurred due to a traffic accident in front of the platoon, during which its maximum deceleration reaches 7.5 m/s².

A. Distributed IDA-PBC Controller

This experiment employs the distributed controller (57), with controller parameters, homogenous/heterogeneous parameters of the CAV platoon specified in Tables I–III, respectively.

In homogeneous case, as position, velocity, and acceleration shown in Fig. 2, the follower vehicles sequentially initiate movement after leader vehicle starts at 10 s, accelerate after 53 s, and execute an emergency brake after 117 s. Under CTH policy, the spacing distance adjusts in real-time according to

the current velocity, which confirms the internal stability stipulated by Corollary 1. These indicate that the follower vehicles can timely track the motion behavior of leader vehicle. As depicted in Fig. 3, the spacing error of each follower vehicle can rapidly converge to zero while maintaining consistency. Furthermore, the level of error gradually diminishes along the backward of platoon, indicating an attenuation property upon string stability, though with a low rate. The peak values of the spacing error during start-up, acceleration, and emergency braking are 7.17×10^{-2} m, 0.111 m, and 0.396 m, respectively.

In heterogeneous case, Fig. 4 indicates that the motion behavior of leader vehicle can still be tracked by follower vehicles. As depicted in Fig. 5, the spacing error still rapidly converges to 0 and maintains a peak level comparable to that in homogeneous case. However, the consistency and attenuation cannot be guaranteed anymore. Without loss of generality, Sections V-B–V-D only consider the heterogeneous case.

B. IDA-PBC Combined With IA

This experiment employs the distributed controller (56), i.e., IDA-PBC controller (57) added with IA (58). The controller parameters and heterogeneous parameters of the CAV platoon are specified in Tables I and III, respectively.

As depicted in Fig. 6, the spacing error peaks at 6.29×10^{-4} m, 1.39×10^{-3} m, and 1.51×10^{-2} m during the start-up,

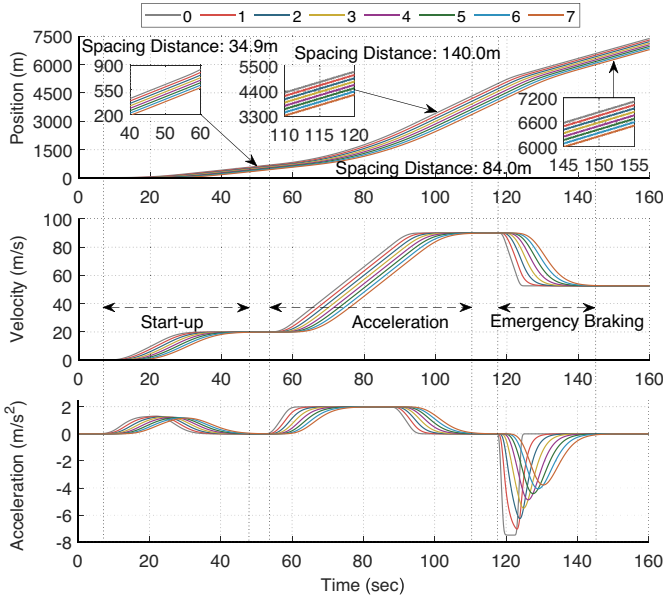


Fig. 2. Motion states of the homogeneous platoon under IDA-PBC.

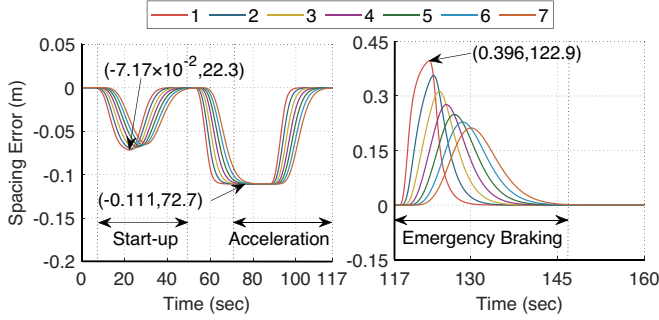


Fig. 3. Spacing error of the homogeneous platoon under IDA-PBC.

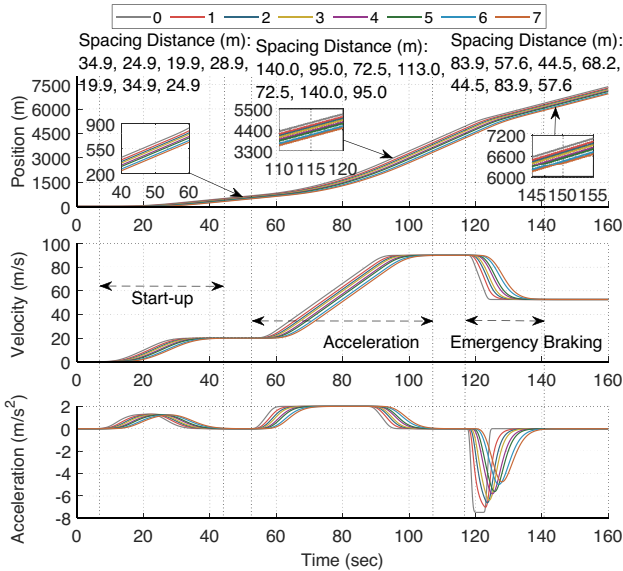


Fig. 4. Motion states of the heterogeneous platoon under IDA-PBC.

acceleration, and emergency braking scenarios, respectively. It is evident that the tracking accuracy and convergence rate have been markedly improved on the addition of IA part,

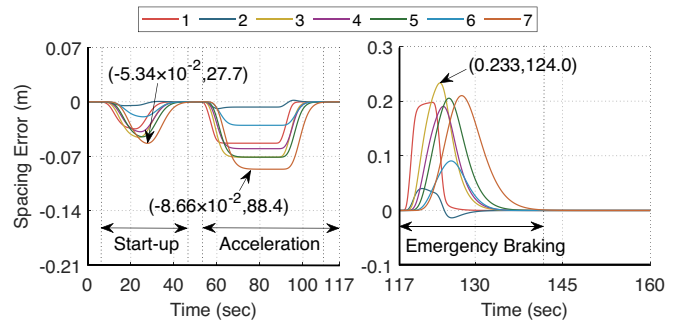


Fig. 5. Spacing error of the heterogeneous platoon under IDA-PBC.

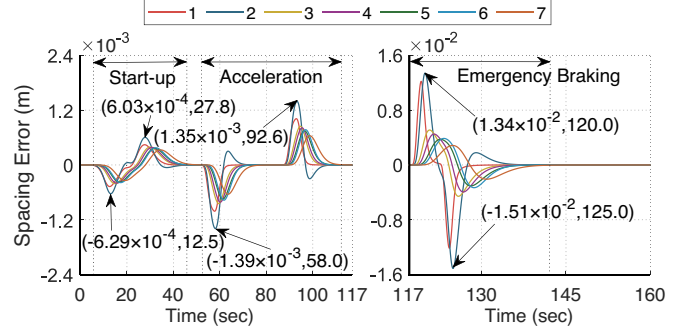


Fig. 6. Spacing error of the heterogeneous platoon under IDA-PBC and IA.

TABLE IV
INTERNAL PARAMETER CONFIGURATION OF MOGA

Internal Parameter	Setting value
Population size n_{pop}	500
Crossover probability	0.8
Mutation probability	0.1
Genome-1 length	20
Genome-2 length	20
Genome-3 length	20
Chromosome length	60
Maximum iteration number G	200
Upper value of fitness $Upper_{\xi,i}$	100
Penalty coefficient $\lambda_{\xi,i}$	100, 100, 100
Attenuation factor η_i	0.95
Sufficient iterations δ_q	20
Parameter bounds	
Θ_{n_i}	[2, 18]
Θ_{j_i}	[2, 18]
$\Theta_{K_{I,i}}$	[1, 12]

compared to the result presented in Fig. 5 where only utilizes IDA-PBC.

C. IDA-PBC Combined With IA and Modified MOGA

The controller (56) is integrated with a modified MOGA in this experiment. The parameters n_i , j_i , and $K_{I,i}$ for the first follower vehicle along with other controller parameters for all follower ones are specified in Table I. The heterogeneous parameters are in accordance with Table III. The internal parameters of modified MOGA are specified in Table IV.

The stable domain corresponding to key controller parameters n_i , j_i , $K_{I,i}$ is determined by penalty term in population-fitness. This can be intuitively filled with the population as the first follower vehicle of which shown in Fig. 7. The Fig. 8 illustrates one online training process

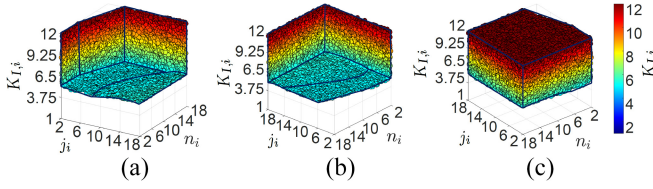


Fig. 7. Stable Domain for the second follower vehicle. (a) Perspective 1. (b) Perspective 2. (c) Perspective 3.

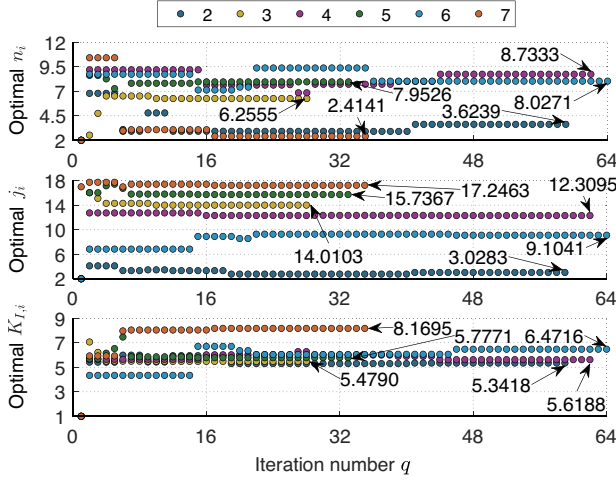


Fig. 8. Training process of n_i , j_i , and $K_{I,i}$.

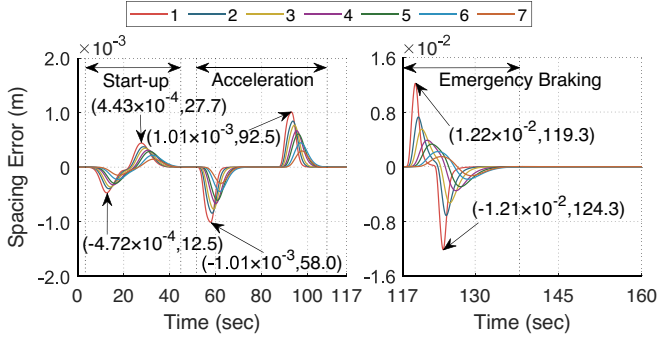


Fig. 9. Spacing error of the heterogeneous platoon under combination of IDA-PBC, IA, and modified MOGA.

of these parameters within stability domain for the second and subsequent following vehicles during a certain training session. It is evident that the population undergoes a certain number of iterations, with the chromosomes gradually stabilizing. Notably, the minimum/maximum number of iterations reaches 28/64. When comparing the results from Fig. 9 with the controller merely combined IDA-PBC with IA in Fig. 6, it is evident that the modified MOGA, while ensuring high-tracking accuracy, rapid convergence, and consistency upon spacing error, equips the controller with ability to handle heterogeneous dynamics and guarantee a stronger attenuation upon string stability than homogeneous case of Fig. 3.

D. Comparison With Existing Approach

This experiment compares the approach in Sections V-B and V-C with an existing approach reported in [15], both of

TABLE V
COEFFICIENTS AND GAINS FOR THE EXISTING APPROACH

$h_i(s)$	c_i	k_i	l_i
0.75	1.00	0.30	6.00
1.00	0.80	0.35	4.80
1.20	0.80	0.45	5.00
1.50	1.00	0.40	4.70

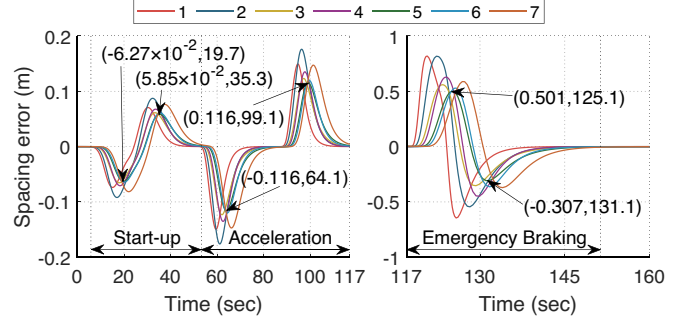


Fig. 10. Spacing error of the heterogeneous platoon under an existing approach.

which are needless of preceding vehicular acceleration. The parameters h_i and τ_i are in accordance with Table III, while the corresponding coefficients c_i and gains k_i , l_i within SMC and DO are configured as per Table V, fulfilling the corresponding string stability criterion.

The trough values depicted in Fig. 10 are 5.85×10^{-2} m, 0.116 m, and 0.307 m during start-up, acceleration, and emergency braking, respectively. The spacing error cannot converged completely at 53, 117, and 150 s in these scenarios and cannot satisfy the attenuation property in heterogeneous case. In contrast, the spacing error in Fig. 6 demonstrates a better tracking accuracy and a more rapid convergence, as the peak values of which is markedly lower than the trough values in existing approach and the spacing error can converge completely at an earlier time. Furthermore, the online training mechanism introduced in Section V-C guarantees a strong attenuation of spacing error via heterogeneous dynamics.

VI. CONCLUSION

This article introduces an intelligent VO-IDA control approach for CAV platoons with heterogeneous dynamics, aiming to ensure both internal stability and string stability. To achieve internal stability, the corresponding problem is translated into tailored tracking control problems for systems with 2 HRD and 1 RD1 channels. By combining backstepping order degradation and virtual stabilization control techniques, VO-IDA prioritizes transforming the PCH form, thus guaranteeing internal stability for CAV platoons. A significant advantage of this proposed controller lies in its ability to indirectly alleviate the dependence on preceding vehicular acceleration without compromising control performance, thereby simplifying implementation and reducing costs for CAV platoons control.

The attenuation of string stability condition is derived using frequency domain analysis methods. A comprehensive

analysis of SFSS under heterogeneous dynamics characterizes the stable domain through a criterion form. Additionally, a linear-proportional relationship among spacing error ratios is identified, providing a foundation for exploring target locations within stable domains. By integrating both aspects, an online MOGA is employed to facilitate this exploration. Simulation results of various control approaches are provided to validate the effectiveness of the proposed control strategy in terms of tracking accuracy, convergence, and its capability to handle heterogeneous vehicular dynamics.

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